

Vehicular Social Dynamic Anomaly Detection With Recurrent Multi-Mask Aggregator Enabled VAE

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Abstract—Vehicle driving behavior analysis and detection tasks have become an indispensable part of intelligent transportation systems. Accurate pattern recognition of potential anomalies during the movement of entities is crucial for improving transportation efficiency. Current methods typically analyze vehicle trajectories independently without considering potential interactions among vehicles. To address this limitation, some studies have integrated graph attention mechanisms to capture the influence of neighboring vehicles during the aggregation process. However, Graph Attention Networks (GATs) are constrained by the univariate nature of attention heads and coefficients, thus lacking flexibility. In this work, we not only consider the social dynamics among neighboring vehicles but also delve into the limitations of GAT models. We propose a Vehicular Social Dynamics Anomaly Detection (VSD-AD) model based on the Recurrent Multi-Mask Aggregator (MMA) enabled Variational AutoEncoder (VAE) architecture to maximize the learning of relational embeddings among neighbors in a highway vehicle network. Furthermore, we apply Node Feature Quantisation (NFQ) to the encoder output to mitigate the complexity of neighbor relationships. Our model is flexible and customizable for different highway scenarios, suitable for large-scale highway vehicle video data. To validate real-world applicability, we further assess its performance on both the simulated dataset and real-world traffic dataset, where our model outperforms other mainstream methods in terms of detection performance.

Index Terms—Anomaly detection, variational autoencoder, graph aggregation, intelligent transportation.

I. INTRODUCTION

HIGHLY autonomous smart traffic management system, as a key component of smart cities, plays a crucial role in addressing urban challenges like traffic flow prediction and accident prevention. With the expansion of urban areas and populations [1], [2] [3], issues such as

traffic congestion and accidents have become more prominent, impacting people's time and safety [4], [5] [6]. Leveraging IoT technologies, such as sensor networks with Road-Side Units (RSUs) and On-Board Units (OBUs), researchers aim to extract valuable insights from data to enhance traffic management systems. Detecting traffic anomalies in a timely manner is essential for efficient traffic management [7], [8] [9], [10], considering the impact of drivers' decisions on traffic dynamics. The challenge for human operators lies in processing vast amounts of data and the lack of a standardized approach to anomaly detection.

In our work, the task is to detect anomalous scenarios in vehicular social dynamics from data collected by highway video surveillance in the sensor network. Vehicle trajectory/behavior anomaly detection tasks can be divided into two major categories based on the data type processed. The first category [11], [12] [13], [14] is devoted to preprocess trajectories extracted from video surveillance systems, then using unsupervised or weakly supervised learning to cluster them into anomalies like sudden braking, severe collision accidents, etc. The second category focuses on spatiotemporal sequences generated from GPS, IoT devices, and motion sensors using methods such as reinforcement learning [15] and deep learning [16], [17] or euclidean space inference [18], with each method concentrating on the operational behavior of individual vehicles.

While the works above have achieved significant results, they have not considered the social dynamics of relationships, as shown in Figure 1, within vehicles and the transmission of messages between drivers. For instance, the brake lights of a car ahead, even if at a considerable distance, can signal the following drivers, who may respond by decelerating. The double flash signals of overtaking warning and alert can prompt the following car to accelerate and prepare to overtake, while the deceleration of the ego-vehicle to give way should not be considered an anomalous behavior. As another example, if vehicles within a driver's forward vision begin to slow down or even stop, the ego-vehicle would naturally decelerate and stop until the road condition becomes positive. In this scenario, the ego-vehicle should not be considered anomalous, but rather the focus should be on the vehicles at the front of the queue. Here, we give a definition of vehicular social dynamic anomalies: in a normally functioning vehicular social network system [19], vehicles form an implied interaction with each other, and if the movement of a vehicle or a number of vehicles negatively affects the normal functioning of the whole system, these vehicles are social dynamic anomalies vehicles.

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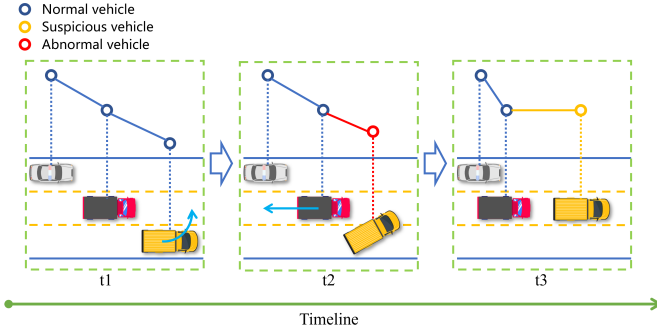


Fig. 1. Example for graph of vehicular social dynamics. Vehicles serve as nodes, and edges are formed when certain conditions are met such as vehicles being too close to each other or engaging in information exchange. Unreasonable lane-changing behavior by vehicles is considered abnormal, and vehicles are flagged as suspected anomalies after completing a lane change.

Some work has noticed this deficiency. Reference [20] captured the interactions between vehicles through the RGAT scheme, but the attention mechanism in GAT is not deep enough and lacks the flexibility of a user-defined model. It can only assign attention coefficients to neighboring vehicles through predefined attention heads. Considering the driving habit differences among drivers from different countries and regions, it is difficult to explore patterns in different data through these methods alone. However, these works have also inspired our approach, as traditional single-vehicle methods are no longer sufficient to accommodate the increasingly complex vehicular social dynamics.

To capture the social nature between vehicles, structuring them into a network-like formation is highly beneficial for vehicle interactions. Graph Neural Networks (GNNs) are powerful tools for dealing with complex relational network structures, and they encompass various specific models such as Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), and Message Passing Neural Networks (MPNNs). These networks, combined with RNNs, can simultaneously process dynamic temporal graphs. However, in designing the network, we still face challenges such as:

- **Incomplete neighbor relationship.** The interaction between drivers and the positional relationships between vehicles often have a significant impact on the driving patterns of vehicles. Therefore, there is still considerable potential for deep exploration of the neighbor relationships between vehicles.
- **Fixed identification patterns.** Due to the varied data distributions across different datasets corresponding to traffic regulations and driving habits in various countries and regions, relying solely on fixed patterns to identify abnormal driving behavior is often inadequate.
- **Inaccurate detection targets.** Many existing approaches have a vague definition of trajectory graphs, leading to coarse detection targets in models. This can result in the loss of some anomalies and ultimately reduce the detection accuracy of the model.

Addressing the challenges outlined above, we propose a data-driven, customizable method for Vehicular Social Dynamic Anomaly Detection (VSD-AD), realized through recurrent multi-mask autoencoders. This method maintains the combination of RNN and graph convolution models to

extract spatiotemporal features of dynamic graphs. It replaces the graph attention mechanism with a more flexible graph multi-mask aggregator to deeply consider the social influence of neighboring vehicles on ego vehicles. To prevent the decoder from being confused by complex encodings from the encoder, inspired by the VQ-VAE model [21], we propose a graph feature quantisation operation to ensure a smooth transition of encoder output representations. In terms of the time dimension, fine-grained sampling is replaced by coarse sampling, enabling the model to detect behavioral anomalies over a longer period. Each sampling period undergoes detailed sampling, i.e., the snapshot of that moment, followed by constructing graphs of vehicles and their neighboring relationships. Furthermore, to facilitate unsupervised learning, we reconstructed the probability distribution of trajectories in the decoder part, highlighting anomalous temporal sequences with lower reconstruction probability.

Fortunately, the proposed model is capable of detecting anomalies in the social dynamics of highway vehicles over extended distances and durations. To validate our model's effectiveness, our data is sourced from large-scale simulation trajectory data generated by the microscopic traffic simulator from [20] and the real-world HighD dataset [22]. Compared with existing state-of-the-art methods, our model is the only one that is customizable and capable of detecting both individual vehicle anomalies and overall scene anomalies within a specific period. Moreover, our method performs robustly with both simulated and real-world data.

The major contributions of this study include:

- We propose a Vehicular Social Dynamic Anomaly Detection (VSD-AD) model based on the Recurrent Multi-Mask Aggregator (MMA) enabled Variational AutoEncoder (VAE) architecture. It captures the social dynamics between highway vehicle neighbors more deeply and offers a degree of customizability.
- We propose a graph vector quantisation technique—Node Feature Quantisation (NFQ) to ensure that the node representations of vehicle relationship graphs aggregated by the encoder are smoothly processed by the decoder.
- We conduct extensive experiments on both simulated and real-world datasets, demonstrating the superior ability of our proposed model to detect anomalous driving behaviors in large-scale highway traffic surveillance systems. Observations on a dataset generated by traffic simulation software demonstrate the model's sensitivity to the social dynamics of vehicles.

II. RELATED WORK

A. Trajectory Modeling

Since trajectory data possess spatiotemporal characteristics, most research on trajectory modeling primarily focuses on learning the spatial features of past trajectories of vehicles, pedestrians, or aerial vehicles and predicting their future movements. An overview of recent years' works in this field can be comprehensively found in [23]. In [24], a grid-based spatial encoder and a relative temporal encoder are combined to extract trajectory features. These features

are then fused with static embeddings (representing static information such as traffic flow and weather) through an attention mechanism. Reference [25] employs network representation learning to achieve accurate vehicle trajectory clustering. Similarly focusing on automobiles, [26] develops three prediction models with distinct architectures: a Transformer-based model (TS-Transformer), a Generative Adversarial Network-based model (TS-GAN), and a Conditional Variational Autoencoder-based model (TS-CVAE). These models demonstrate that traffic state-based models consistently predict future trajectories more accurately than vanilla models, with the TS-Transformer producing state-of-the-art results. Yan et al. [27] proposes an unsupervised method for traffic VAD based on future object localization. In the field of aircraft, the learning model of [28] adopts an encoder-decoder architecture with a hybrid core of LSTM and GAN. Li et al. [29] proposes the traffic trajectory prediction algorithm based on the convolutional attention network (TraGCAN) to predict the trajectories of heterogeneous traffic agents in dense traffic. They construct the spatial relationship of traffic agents as a graph structure and introduce a graph convolutional network to extract spatial interactions. Reference [30] utilizes Graph Convolutional Networks (GCNs) to design modules for capturing sub-trajectory anomalies in both attribute space and structural space within the generation and contrastive learning modules.

Our work involves capturing all trajectory points within a fixed area in a snapshot manner and recording vehicle information for each point, including position, velocity, and acceleration. We have designed an encoder-decoder framework to learn the spatiotemporal relationships among vehicles in these snapshots, with the aim of modeling these spatiotemporal relationships in trajectories.

B. Graph Anomaly Detection

In this paper, we focus on modeling the relationships between vehicles and their neighbors on a highway scene as a graph structure, with the fundamental task being graph anomaly detection. Relevant reviews can be found in [31] and [32], both of which are classic works on using deep learning methods to solve graph anomaly detection problems. The definition of anomalies varies, but common criteria for judgment include extreme traffic events and accidents [33], driving behaviors [34], [35], and autonomous vehicle driving conditions [36]. Researchers have developed a plethora of graph-based methods for these detection scenarios. In the article [37], Deng proposes a spatiotemporal graph convolutional adversarial network (STGAN). Other approaches include using GNN methods for multivariate time-series anomaly detection [38], [39], and generative and contrastive self-supervised learning methods [40]. Tu et al. [41] introduced Deep outdated fAct detectionN (DEAN), which employs a Relations-to-Nodes (R2N) graph, a contrastive approach designed to unveil implicit outdated information. CenGCN [42] posits that the information transmitted from other nodes to the core node through the propagation mechanism is unequal, a concept that bears resemblance to the Graph Attention Network (GAT) [43]. Chen et al. [44] combines sequence and graph-structured data, achieving sequence reconstruction through an

attention-based graph convolutional network. Wang et al. [45] proposed a method integrating reinforcement learning with spatial knowledge graphs to predict human mobility patterns. Notably, [46] focuses on improving the generalizability of graph anomaly detection models via data augmentation. There are also numerous methods targeting subgraphs, such as the Anomalous Subgraph Autoencoder (AS-GAE) proposed by [47] to extract anomalous subgraphs. Reference [48] performs intra-view node-subgraph contrastive learning to enrich the anomaly information. Reference [49] starts from injective node augmentation and tries to enrich node representations via multi-scale contrastive training. As the most relevant work, [20] differs from our adoption of MMA. Their proposed model utilizes a GAT to extract vehicle neighborhood relationships. However, due to insufficient depth in capturing spatiotemporal relationships with the GAT and GRU combination, the model lacks expressive power. This is the primary challenge addressed by our proposed method in this study. Additionally, we introduce node quantization to further fine-tune the model.

Inspired by anomaly detection in graphs, we model each snapshot as a graph, where vehicles are nodes and the adjacency relationships between vehicles are edges. Thus, we transform the problem of detecting anomalies in vehicle trajectories into a graph node anomaly detection problem.

III. METHOD

In this section, we introduce the proposed VSD-AD model which consists of three parts, namely Encoder module, Decoder module, and NFQ module. The overview of VSD-AD can refer to Figure 2.

A. Problem Formulation

The input to the model consists of an observation set $O = \{o_1, o_2, \dots, o_{|\mathcal{W}|}\}$ over a time window $\mathcal{W}$ on the highway. For each time step t within $\mathcal{W}$, the observation $o_t = \{x_t^1, x_t^2, \dots, x_t^N\}$ is noted, where $N = |o_t|$ represents the number of vehicles in the current observation. Here, x_t^i denotes the features of the vehicle numbered i at time step t . The vehicle features $x_t^i = (x_t^i, y_t^i, l_t^i, v_t^i, a_t^i)$ include the longitudinal position x_t^i , lateral position y_t^i , driving lane ID l_t^i , longitudinal speed v_t^i , and longitudinal acceleration a_t^i . Furthermore, to reduce feature complexity and training difficulty, we opt not to include vertical vehicle velocity and acceleration as features in training. This decision stems from the observation that in highway datasets, the majority of vehicles exhibit predominantly horizontal velocities and accelerations, with minimal values in the vertical direction. Even during turns, the deviations in the vertical direction are minimal. In the rare event of abrupt maneuvers, such as sharp turns, the model can effectively capture and learn from the significant decreases in horizontal velocity and acceleration values.

Given the vehicle features x_t^i for all t in $\mathcal{W}$, our primary objective is to detect vehicles exhibiting anomalous social behaviors over a certain period $\mathcal{W}$ on the highway section. Additionally, we will segment the highway section into shorter stretches of length δ_s , referred to as scenes, and perform anomaly detection for all vehicles on stretch S during $\mathcal{W}$.

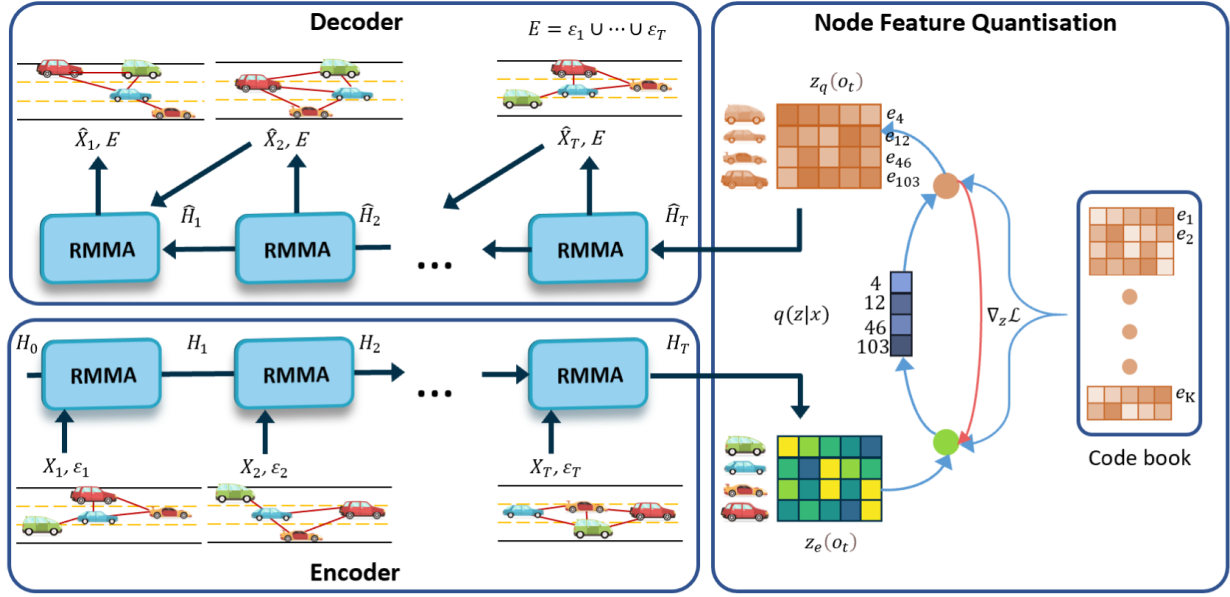


Fig. 2. The overview of VSD-AD model which consists of three modules: Encoder, Decoder, and Node Feature Quantisation. Firstly, within a time window, the snapshot of road vehicles composing the graph is mined for neighbor relationships using Recurrent Multi-Mask Aggregators. Then, the output of the encoder is quantized by the Node Feature Quantization module. Finally, reconstruction is performed by a decoder with a structure identical to that of the encoder.

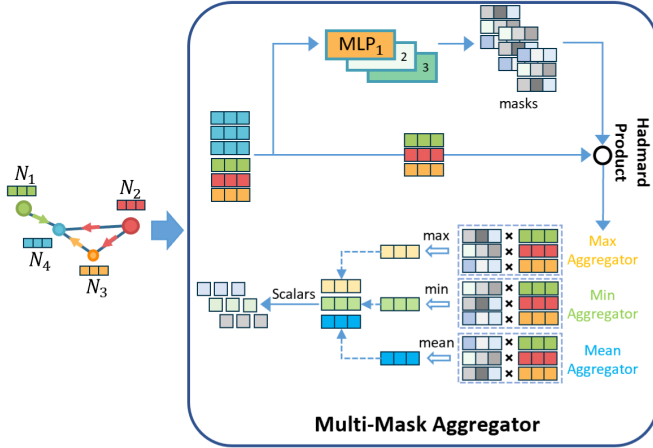


Fig. 3. The Multi-Mask Aggregator within the GRU module will learn masks for each vehicle node feature using MLP (Multi-Layer Perceptron) and aggregate them onto ego vehicle nodes through various aggregators.

The criterion for scene anomaly detection involves identifying anomalous vehicles within each segment, as elaborated in Section III-G.

B. Graph Construction

We first construct a spatiotemporal dynamic graph representing the vehicular social dynamics, denoted as $G(\mathcal{W}) = \{(\mathcal{V}_1, \mathcal{E}_1), (\mathcal{V}_2, \mathcal{E}_2), \dots, (\mathcal{V}_T, \mathcal{E}_T)\}$ for time t in $\mathcal{W} = \{1, 2, \dots, T\}$. The node set $\mathcal{V}_t$ represents all vehicles observed on the highway at o_t and their features $x_t^i = (x_t^i, y_t^i, l_t^i, v_t^i, a_t^i)$. From the formulation, it is evident that for different time steps t in $\mathcal{W}$, the number of vehicles $N = |\mathcal{V}_t|$ is fixed, but the number of vehicles at a single time step $|o_t|$ can vary, similar to the count in a snapshot. For e_t^{ij} in $\mathcal{E}_t$, we define its existence based on the condition that, centered on the current ego vehicle x_t^i , vehicle x_t^j is within δ_x feet horizontally and less than δ_l lanes vertically. This approach is justified as, in real scenarios,

drivers primarily focus on the few vehicles closest to them, and the most significant impact on a current vehicle is undoubtedly from its neighboring vehicles.

C. Encoder

In this subsection, we introduce how Recurrent Multi-Mask Aggregators [50] are used to compress and encode trajectory time series data.

Specifically, we employ the RNN variant GRU [51], known for its proficiency with sequential data tasks. Compared to the traditional LSTM (Long Short-Term Memory network), GRU's structure is simpler. It merges the forget and input gates into a single "update gate" and omits the output gate. This simplified structure results in fewer parameters for the GRU, making it more computationally efficient. We replace the matrix multiplication in the GRU's update gate z_t , reset gate r_t , and candidate state $\tilde{H}_t$ with graph convolution operations, denoted as $*_{\mathcal{M}}$, to capture the social dynamics among vehicle graphs. This approach effectively leverages the strengths of GRU in handling temporal dependencies while adapting its architecture to better represent the complex social dynamics present in vehicular graphs:

$$\begin{aligned} z_t &= \sigma(\mathbf{W}_{xz} *_{\mathcal{M}} \mathbf{X}_t + \mathbf{W}_{hz} *_{\mathcal{M}} \mathbf{H}_{t-1} + \mathbf{b}_z), \\ r_t &= \sigma(\mathbf{W}_{xr} *_{\mathcal{M}} \mathbf{X}_t + \mathbf{W}_{hr} *_{\mathcal{M}} \mathbf{H}_{t-1} + \mathbf{b}_r), \\ \tilde{H}_t &= \tanh(\mathbf{W}_{xh} *_{\mathcal{M}} \mathbf{X}_t + \mathbf{W}_{hh} *_{\mathcal{M}} (r_t \odot \mathbf{H}_{t-1} + \mathbf{b}_h), \\ \mathbf{H}_t &= z_t \odot \mathbf{H}_{t-1} + (1 - z_t) \odot \tilde{H}_t. \end{aligned} \quad (1)$$

where $H_t \in \mathbb{R}^{N \times d_h}$ represents the hidden state, while $z_t \in \mathbb{R}^{N \times d_h}$ and $r_t \in \mathbb{R}^{N \times d_h}$ correspond to the update gate and reset gate, respectively, each with a dimension of d_h . The weights W_{xz} , W_{hz} , W_{xr} , W_{hr} , W_{xh} , W_{hh} and biases b_z , b_r , b_h are learnable parameters. The activation function σ is the sigmoid function, and $\odot$ denotes element-wise multiplication. The input $X_t \in \mathbb{R}^{N \times d_f}$ at time step t to the GRU comes

from the raw observation o_t , where d_f is the node feature dimension. This dimensionality is derived from the spatial coordinates, speed, and acceleration $(x_t, y_t, v_t, a_t) \in \mathbb{R}^{N \times 4}$, concatenated with lane embeddings $h_{l_t} \in \mathbb{R}^{N \times d_l}$. It is important to note that h_{l_t} represents the mapping of discrete lane IDs to their corresponding embedding vectors with dimension d_l , making the input feature dimension $d_f = 4 + d_l$.

For the graph convolution operation $*_{\mathcal{M}}$, unlike the RGAT method based on Graph Attention Network (GAT) [43] used in [20] to capture dynamic interactions between neighboring vehicles, we adopt RMMA [50] based on Multi-mask aggregators. This approach draws similarities with GAT and MPNN but offers some theoretical and practical advantages [50]. As illustrated in Figure 3, the graph convolution operation $*_{\mathcal{M}}X_t$ for the vehicle feature set input X_t primarily consists of four processes:

a. *Mask Generation*. The purpose of this step is to learn unique values between node neighbors, forming a mask. The Multi-Mask Aggregators (MMA) deploy a Multi-Layer Perceptron (MLP) between each node pair x^i, x^j and selectively extract node features e^{ij} . This operation can be likened to creating weights for multiplying aggregated neighboring features, similar to the messages passed in Message Passing Neural Networks (MPNN) and the attention coefficients in Graph Attention Networks (GAT). However, while GAT typically yields a single attention coefficient, the mask in MMA can learn a relationship vector for each pair of neighbors. For each node pair x^i, x^j , the learned mask is represented as:

$$m_t^j = \text{MLP}(\|x_t^i, x_t^j, e_t^{ij}\rangle) = \sigma(W_m(\|x_t^i, x_t^j, e_t^{ij}\rangle)) \quad (2)$$

where m_t^j represents the mask for vehicle j at time step t , σ is the activation function and W_m is a learnable matrix. The symbol $\|$ denotes column-wise concatenation. Since $x_t^i, x_t^j, e_t^{ij} \in \mathbb{R}^{d_f}$, the result of the concatenation belongs to $\mathbb{R}^{3 \times d_f}$, and $W_m \in \mathbb{R}^{3 \times \alpha}$. The dimensionality of the result of multiplying the concatenated x_t^i, x_t^j, e_t^{ij} with W_m is in $\mathbb{R}^{d_f \times \alpha}$, which ultimately determines the dimension of m_t^j . Here, α represents the hyperparameter for the number of hidden units.

b. *Neighbor Aggregation*. Unlike most graph convolution methods that use only one type of aggregator, such as a mean aggregator, to combine neighbor information, Multi-Mask Aggregators (MMA) allow users to choose one or multiple masked aggregators based on the data distribution and their specific needs. This flexibility enables a focus on the social dynamics of vehicle interactions from multiple perspectives. There are three types of aggregators available, namely masked max, min, and mean aggregators:

$$\max_t^j = \max_{j \in \text{Nei}_t^i}(x_t^j \circ m_t^j), \quad (3)$$

$$\min_t^j = \min_{j \in \text{Nei}_t^i}(x_t^j \circ m_t^j), \quad (4)$$

$$\text{mean}_t^j = \frac{1}{|\text{Nei}_t^i|} \sum_{j \in \text{Nei}_t^i} x_t^j \circ m_t^j, \quad (5)$$

where Nei_t^i denotes the set of neighbor IDs for vehicle i at time step t . Each aggregator performs a Hadamard product with the neighbor features and the masks generated in step a). The difference lies in the operations of min and max, which select the smallest or largest vector from the result vectors for

aggregation. This allows the network to learn to ignore certain “undesired” nodes when propagating information. The mean aggregator, on the other hand, integrates information from all neighbor vectors, where $\circ$ denotes the Hadamard product operation.

c. *Message Vector Scaling*. Further, enhances expressiveness by utilizing degree scalars to amplify and attenuate signals based on the nodes’ degrees. The general expression for degree scalars $S(\cdot)$ is:

$$S(d, \omega) = \left(\frac{\log(d+1)}{\varphi} \right)^\omega, \quad d > 0, -1 < \omega < 1, \quad (6)$$

in this context, ω represents the amplification factor, where d is the node degree, and φ is the average degree of the training set. In our work, we set $\omega = \{-1, 0, 1\}$, corresponding to signal attenuation, no change, and amplification, respectively.

d. *Assimilation*. Finally, it is necessary to integrate the results of all aggregators and scalars. If $\otimes$ denotes the tensor product, then the universal aggregation function $\oplus_{\text{mask}}$ for MMA can be expressed as:

$$\oplus_{\text{mask}} = \begin{pmatrix} I \\ S(D, \alpha = 1) \\ S(D, \alpha = -1) \end{pmatrix} \otimes \begin{pmatrix} \text{Masked Max} \\ \text{Masked Min} \\ \text{Masked Mean} \end{pmatrix}. \quad (7)$$

At this point, for each time step t , all vehicles x_t^i in the observation set o_t have received information from their neighbors. Information that cannot influence the ego vehicle will be ‘discarded’, while neighbor information that truly alters driving behavior will be strengthened. In comparison to works with predefined edge weights, it is evident that MMA exhibits stronger interpretability and possesses more learnable parameters to determine the social dynamics among neighbors than GAT. Moreover, the multi-aggregator feature of MMA provides it with a certain degree of flexibility and customization. For instance, operators can choose or combine multiple min aggregators as needed.

D. Node Feature Quantisation

Inspired by Vector Quantized-Variational AutoEncoder (VQ-VAE) [21], which has proven effectiveness in vector quantisation when the posterior and prior distributions are categorical, the VQ operation has been shown to improve the overall network performance by drawing an embedding table from these distributions and using it as input to the Decoder network. In this subsection, we will provide a detailed explanation of our proposed Node Feature Quantisation (NFQ) mechanism.

The input to NFQ is the hidden state of the Encoder’s last layer, denoted as $H_T \in \mathbb{R}^{N \times d_h}$. First, we define a discrete latent embedding space $e \in \mathbb{R}^{K \times d_h}$, where K is the size of the discrete latent space. Thus, there are K embedding vectors $e_i \in \mathbb{R}^{d_h}$, $i \in 1, 2, \dots, K$, constituting the embedding space known as the code book. Our goal is to map each vector in H_T to the code book e for quantisation. The initialization methods for a code book typically include random initialization and K-means clustering initialization. Here, we initialize each vector in the code book by sampling each element from a standard normal distribution. We then compute the posterior

categorical distribution probabilities $q(z|o_t)$, which are defined as one-hot vectors, as follows:

$$q(z = k|o_t) = \begin{cases} 1 & \text{for } k = \operatorname{argmin}_j \|z_e(o_t) - e_j\|_2, \\ 0 & \text{otherwise.} \end{cases} \quad (8)$$

the variable $z_e(o_t)$, denoted as H_T , represents the result of processing the original data o_t through the Encoder. The distribution $q(z = k|o_t)$ is deterministic and is based on the principle of computing the distances between each input vector and the K vectors in the embedding. The distances are calculated using the common Euclidean distance $\|\cdot\|_2$. The result of Equation (8), $q(z = k|o_t) \in \mathbb{R}^{N \times K}$, indicates that only the vectors corresponding to $z_e(o_t)$ with the closest Euclidean distances to the e_i vectors in the code book e will have the value 1 in the item at index i of the $q(z = k|o_t)$ vector. all other values are set to 0. The quantized result $z_q(o_t)$ can be expressed as:

$$z_q(o_t) = q(z = k|o_t) \odot e. \quad (9)$$

the described process can be summarized as representing $z_e(o_t)$ through a discrete bottleneck and then mapping it to the nearest elements in the embedding e :

$$z_q(o_t) = e_k, \quad \text{where } k = \operatorname{argmin}_j \|z_e(o_t) - e_j\|_2. \quad (10)$$

Due to the fact that messages from multiple neighbors are aggregated into node tensors by multiple aggregators through Recurrent Multi-Mask Aggregation (RMMA), and these tensors are then passed as hidden states to the next RMMA, the information content tends to be coarse. We believe this may be detrimental to the subsequent tasks of data reconstruction and classification by the Encoder. Therefore, by employing the proposed Node Feature Quantisation (NFQ), we aim to map the output of the Encoder to discretized entries, introducing a quantisation operation. This operation is intended to help alleviate the aforementioned issue.

E. Decoder

The Decoder employs the same RMMA structure as the Encoder. Furthermore, both the Decoder and Encoder have an equal number of spatiotemporal modules in their RMMA structures, determined by the depth of the input spatiotemporal dynamic graph, i.e., our window size $\mathcal{W}$. To avoid the computational burden of repeatedly calculating the edge set, we use the union of all edges in the Encoder, denoted as $E = \mathcal{E}_1 \cup \mathcal{E}_2 \cup \dots \cup \mathcal{E}_T$, as the input for the Decoder. According to our graph construction approach, the edge set only restricts the neighborhood around the nodes. The interaction relationships between neighboring nodes and which neighbors have a more significant impact on the central node are dynamically determined by the multi-mask aggregation mechanism of RMMA.

The hidden state $\hat{H}_t \in \mathbb{R}^{N \times d_h}$ of the Decoder is further processed through a fully connected layer to generate the output node state reconstruction $\hat{X}_t$. Details about $\hat{X}_t$ will be explained in subsection F. Subsequently, the output $\hat{X}_t$ serves as the input for the next recurrent step. Unlike the Encoder, since the vectors at time T are most relevant to the state at

time T , we decode the sequence backward from time step T . Therefore, the initial input for the Decoder-RMMA will be the node states at time T .

F. Loss Function

For each ego vehicle at time t , certain features including the longitudinal position x_t^i , velocity v_t^i , and acceleration a_t^i are assumed to follow univariate Gaussian distributions, i.e., $x_t^i \sim \mathcal{N}(\mu_{x_t^i}, \sigma_{x_t^i})$, $v_t^i \sim \mathcal{N}(\mu_{v_t^i}, \sigma_{v_t^i})$, and $a_t^i \sim \mathcal{N}(\mu_{a_t^i}, \sigma_{a_t^i})$. Since lane information is a discrete signal, the longitudinal lane information l_t^i is presumed to be a discrete choice among several lanes L . For the Decoder, the ego vehicle's lane position is a potential discrete category distribution, represented as $p_{l_t^i}, \dots, p_{L_t^i}$. The Decoder's output $\hat{X}$ will eventually evaluate the above Gaussian and categorical distributions, meaning that for each $\hat{x}^i \in \hat{X}$, there are $\hat{x}^i = [\hat{\mu}_{x_t^i}, \hat{\sigma}_{x_t^i}, \hat{\mu}_{v_t^i}, \hat{\sigma}_{v_t^i}, \hat{\mu}_{a_t^i}, \hat{\sigma}_{a_t^i}, \hat{p}_{l_t^i}, \dots, \hat{p}_{L_t^i}]$ to evaluate the mean, variance of the longitudinal position, velocity, acceleration, and the probability of the lane position.

The probability density functions for evaluating position, velocity, and acceleration are denoted as $q(x_t^i|\hat{\mu}_{x_t^i}, \hat{\sigma}_{x_t^i})$, $q(v_t^i|\hat{\mu}_{v_t^i}, \hat{\sigma}_{v_t^i})$, and $q(a_t^i|\hat{\mu}_{a_t^i}, \hat{\sigma}_{a_t^i})$ respectively. We aim to minimize the negative log-likelihoods as follows:

$$\mathcal{L}_{x_t^i} = -\log \left(q \left(x_t^i | \hat{\mu}_{x_t^i}, \hat{\sigma}_{x_t^i} \right) \right), \quad (11)$$

$$\mathcal{L}_{v_t^i} = -\log \left(q \left(v_t^i | \hat{\mu}_{v_t^i}, \hat{\sigma}_{v_t^i} \right) \right), \quad (12)$$

$$\mathcal{L}_{a_t^i} = -\log \left(q \left(a_t^i | \hat{\mu}_{a_t^i}, \hat{\sigma}_{a_t^i} \right) \right). \quad (13)$$

For lane classification, we aim to minimize the cross-entropy loss as follows:

$$\mathcal{L}_{l_t^i} = -\sum_{l=1}^L \mathbb{1}_{l_t^i} \log \left(\hat{p}_{l_t^i} \right), \quad (14)$$

where $\mathbb{1}_{l_t^i}$ is an indicator function that equals 1 if vehicle $\mathbf{x}_t^i$ is in lane l at time t , and 0 otherwise.

Given the direct gradient estimation from $z_q(o_t)$ to $z_e(o_t)$ and that the above loss functions do not account for the gradient descent problem for each embedding e_i in Node Feature Quantisation (NFQ), to learn the code book's embedding space, we employ an l_2 error to approximate the Encoder outputs $z_e(o_t)$ towards embedding vectors e_i . The loss function for learning the code book, $\mathcal{L}_{NFQ}$, is defined to minimize the l_2 error as follows:

$$\mathcal{L}_{NFQ} = \| \operatorname{sg}[z_e(o_t)] - z_q(o_t) \|_2^2 + \| z_e(o_t) - \operatorname{sg}[z_q(o_t)] \|_2^2, \quad (15)$$

where $\|\cdot\|_2$ denotes the Euclidean distance. $\operatorname{sg}[\cdot]$ represents the stop gradient operator, which is defined as the identity function during forward computation and has zero partial derivatives, effectively treating its operand as an un-updated constant. Thus, the overall loss function can be expressed as a weighted sum of the negative log-likelihood losses and the cross-entropy loss across all agents and all times, as well as the code book's l_2 error:

$$\mathcal{L}_t^i = \lambda_x \mathcal{L}_{x_t^i} + \lambda_v \mathcal{L}_{v_t^i} + \lambda_a \mathcal{L}_{a_t^i} + \lambda_l \mathcal{L}_{l_t^i}, \quad (16)$$

$$\mathcal{L} = \sum_{i=1}^N \sum_{t=1}^T \mathcal{L}_t^i + \lambda_{NFQ} \mathcal{L}_{NFQ}. \quad (17)$$

where λ_x , λ_v , λ_a , λ_l , and λ_{NFQ} are the weights for each loss component, serving as hyperparameters. Typically, these can be empirically set as $\lambda_x = 1$, $\lambda_v = 1$, $\lambda_a = 2$, $\lambda_l = 2$ and $\lambda_{NFQ} = 2$.

G. Anomaly Detection

Based on the granularity of detection, anomaly detection can be divided into anomaly *vehicle* detection and anomaly *scene* detection.

For anomaly vehicle detection, the anomaly score $\rho_{\mathcal{W}}^i$ of the vehicle i under time window $\mathcal{W}$ is calculated as its average loss across all time steps:

$$\rho_{\mathcal{W}}^i = \frac{1}{T} \sum_{t=1}^T \mathcal{L}_t^i, \quad (18)$$

where T is the length of the time window $\mathcal{W}$. Anomaly scene detection primarily identifies anomalies in the entire scene, aggregating the losses of all vehicles within stretch $\mathcal{S}$ during $\mathcal{W}$ to calculate the scene anomaly score $\rho_{\mathcal{W}}^{\mathcal{S}}$:

$$\rho_{\mathcal{W}}^{\mathcal{S}} = \max(\mathcal{L}_t^i), \forall (i, t) \text{ that } x_t^i \in \mathcal{S}. \quad (19)$$

The equation indicates that each scene's anomaly score is determined by the maximum anomaly score among its vehicles, implying that the presence of any anomalous vehicle within a scene renders the entire scene abnormal. Here, we obtain the set of anomaly scores *VeSco* for vehicles and the set of anomaly scores *SceSco* for scenes.

After obtaining these anomaly scores, we cannot directly determine anomalous vehicles and scenarios. However, we can establish thresholds based on the intuitive distribution of these anomaly scores. To ensure scientific rigor, this study employs cross-validation by splitting the dataset into training and validation sets. Through ROC curve analysis on the validation set, we select the threshold that maximizes the Area Under the Curve (AUC). This method allows for a more precise threshold selection, suitable for scenarios that require a balance between prediction accuracy and recall.

It is worth mentioning that, unlike using the mean for aggregation, we adopt maximum aggregation to highlight the presence of anomalies, rather than averaging the losses of normal vehicles. In conclusion, the overall process and pseudo code of the VSD-AD model proposed by us can be referenced in Algorithm 1.

IV. EXPERIMENTS

In this section, we utilize two types of data sources, namely simulation data and real-world data, to validate the performance of our proposed VSD-AD model. Initially, simulation data with ground truth labels for anomalies is used for a quantitative performance comparison between our proposed model and baseline models. Subsequently, the HighD dataset is employed to verify the effectiveness of our model in practical applications. Moreover, to demonstrate the strong customization capability of VSD-AD, we modified different modules of the VSD-AD model on both datasets to fit actual needs.

Algorithm 1 The Overall Procedure of VSD-AD

Require: Scenario: *sce*, Training set: D_{train} , Test set: D_{test} , Training epochs: *epoches*, Batch size: *batch_size*, Window size: $\mathcal{W}$.

Ensure: Anomaly scores ρ of trajectory test set.

- 1: Construct vehicular social dynamics graph $G(\mathcal{W})$ for D_{train} and D_{test} , as shown in Section III-B.
- 2: Initialize the trainable parameters W and the code book of feature quantisation e randomly.
- 3: // *Training phase*
- 4: **for** *epoch* in $\{1, 2, \dots, epoches\}$ **do**
- 5: **for** batch ($G(\mathcal{W})_1, G(\mathcal{W})_2, \dots, G(\mathcal{W})_{batch_size}$) **do**
- 6: **for** t in $\{1, 2, \dots, T\}$ **do**
- 7: Construct lane embeddings h_l .
- 8: Get node features X_t via concating with h_l .
- 9: Get hidden state H_t via Eq. (1).
- 10: **end for**
- 11: Compute $q(z|o_t)$ via Eq. (8).
- 12: Get the quantized result $z_q(o_t)$ via Eq. (9)
- 13: **for** t in $\{1, 2, \dots, T\}$ **do**
- 14: Obtain hidden state $\hat{H}_t$ via Eq. (1).
- 15: **end for**
- 16: Compute the position $\mathcal{L}_x$, velocity $\mathcal{L}_v$, acceleration $\mathcal{L}_a$ and lane loss $\mathcal{L}_l$ via Eq. (11) (12) (13) (14).
- 17: Compute the NFQ loss via Eq. (15).
- 18: Combine all the loss as $\mathcal{L}$ via Eq. (16) (17)
- 19: Optimize $\mathcal{L}$, update parameters W and code book e .
- 20: **end for**
- 21: **end for**
- 22: // *Test phase*
- 23: **for** o_t in D_{test} **do**
- 24: Obtain the loss $\mathcal{L}_t^i$ for cars via Eq. (16).
- 25: Obtain the anomaly *car* score $\rho_{\mathcal{W}}^i$ via Eq. (18).
- 26: Obtain the anomaly *scene* score $\rho_{\mathcal{W}}^{\mathcal{S}}$ via Eq. (19).
- 27: Obtain the anomaly scores set *VeSco* and *SceSco*.
- 28: **return** anomaly scores *VeSco* and *SceSco*.
- 29: **end for**

A. Datasets

The detailed information for both datasets is presented as follows:

1) *TransModeler Simulation Dataset*: This dataset originates from the virtual simulation data generated by TransModeler, as referenced in [20]. TransModeler, a traffic simulation software, accommodates a wide range of granularity and complexity in simulation needs and generates traffic flow on roads and intersections, including vehicle acceleration, deceleration, and lane-changing behaviors. The data for this work were captured over a 5-mile stretch of a 4-lane highway at a frequency of 1Hz, encompassing various scenarios and vehicle behaviors. Detailed information can be found in Table I. In Table I, the first column represents the scenario types, and the first row indicates the types of vehicles present in each scenario. "CTG" is the abbreviation of "constant time gap", which stands for the vehicle that follows the constant time car-following model where drivers can maintain a consistent desired distance from the vehicle in front. The "TD" column, representing

TABLE I
DISTRIBUTION OF VEHICLE TYPES IN DIFFERENT SCENARIOS

	Normal	Speeding	Slow	CTG	Stalled	TD
Normal	90%	5%	5%	-	-	500-1600
Speeding	70%	30%	-	-	-	500
Slow	98%	-	2%	-	-	500
tailgating	53.20%	-	-	46.80%	-	500
Stalled car	other	-	-	-	15	1500
Comprehensive	89%	10%	1%	46.80%	2	1000

TABLE II
SIMULATION DATA TRAINING SET AND TEST SET DISTRIBUTION

	Scenario	Car count	Trajectory count
Simulation	Training	7886	920311
	Comprehensive	1281	147254
	Slow	636	75809
	Speeding	583	65685
	tailgating	645	78600
	stalled	1922	263370

TABLE III
HIGHD DATA TRAINING SET AND TEST SET DISTRIBUTION

	Scenario	Car count	Trajectory count
HighD	Training	42106	459187
	Testing	1795	21840

“Traffic Demand”, displays the specific traffic volume for each scenario in terms of vehicles per lane per hour, simulating various traffic demands including free flow and congested conditions. The remaining values in the table indicate the distribution of each vehicle type within the respective scenario. The dataset is divided into six distinct scenarios. Among them, the Comprehensive scenario consists predominantly of normal vehicles along with all other types of abnormal vehicles. This scenario is complex and closely reflects real-world conditions, thereby emphasizing the model’s robustness. In the Normal scenario, abnormalities are limited to Speeding and Slow categories only. Each of the remaining scenarios focuses on a single abnormality type, highlighting the common limitation of models in detecting only one or two specific anomalies rather than a comprehensive range.

Unlike most studies that focus only on common anomalies such as speeding, tailgating, and stalled vehicles, we have also examined slow driving behaviors, which are difficult to learn using single-vehicle methods. The specific distribution of the simulation dataset in this work is presented in Table II.

2) *HighD Dataset*: The real-world dataset HighD was collected by the Technical University of Munich using drones and other tools on local highways, amassing a total of 16.5 hours of high-definition traffic video data. In this work, we selected approximately 5.8 hours of highway vehicle trajectory data as our training set, which includes scenarios of both clear and congested traffic conditions on a three-lane highway, with corresponding traffic flows ranging from 1200 to 3600 vehicles per lane per hour. Subsequently, we used a 15-minute segment with a traffic flow of approximately 2300 vehicles per lane per hour as our test set. The detailed distribution is presented in Table III.

B. Baselines and Metrics

1) *Baselines*: To demonstrate the superiority of our work, we compared it with the most advanced trajectory anomaly detection algorithms and heuristic algorithms on the same trajectory dataset. Our baselines primarily include:

- **Linear Temporal Interpolation (LTI)** [52] approximates a trajectory by interpolating between trajectory points at the first and last time steps.
- **Constant Velocity Model (CVM)** [53], which assumes that an agent’s next step velocity and direction are the same as the previous two sampling points. This approach has even surpassed some neural network algorithms due to their erroneous assumptions.
- **Robust Tensor Recovery (RTR)** [54] introduces a tensor decomposition method that helps recover low-rank tensor values containing vehicle information and compares them with normal patterns to detect sparse anomalies.
- **Sequence to Sequence Learning (seq2seq)** [55] employs a classic encoder-decoder neural network architecture with LSTM to capture the spatiotemporal relationships of vehicle trajectories, which is used to recognize patterns.
- **Spatiotemporal Graph Auto-Encoder (STGAE-biv)**, a variant from [52], defines a weighted adjacency matrix through a kernel function incorporating the distances between agents and uses a bivariate Gaussian reconstruction loss to build the encoder-decoder to obtain the anomaly score.
- **DSAB** [20], an autoencoder composed of RGAT blocks that captures the relationships of social dynamics between vehicles, where the most crucial vehicles for the ego-vehicle are assigned attention coefficients through an attention mechanism.

Among all the baselines, LTI and CVM are simple parameter-free reconstruction methods, while the others require learning model parameters to capture non-linear patterns. Both can only handle simple non-free-flow scenarios with minor vehicle speed changes. RTR, Seq2seq, CVM, and LTI consider only the trajectory of the ego-vehicle itself, while STGAE-biv and DSAB demonstrate the social dynamics interactions between vehicles.

2) *Metrics*: Similar to the validation metrics used in most anomaly detection works in practical experiments, we adhere to the following metrics:

- **ROC-AUC score**, a common indicator in the field of anomaly detection. Firstly, sort real labels and Sco based on the descending order of Sco values. Subsequently, it calculates the AUC using the sorted Sco and corresponding real labels’ values. The ROC curve is plotted with False Positive Rate (FPR) on the x-axis and True Positive Rate (TPR) on the y-axis. The AUC under the ROC curve is typically computed using the following formula:

$$AUC = \frac{\sum_{i \in pos} \sum_{j \in neg} \mathbb{1}(p_i > q_j)}{|pos| \cdot |neg|}, \quad (20)$$

where pos and neg are sets of positive and negative class samples respectively, p_i and q_j represent the predicted probabilities (here denoted as Sco) for the i -th

positive sample and j -th negative sample. The symbol $\mathbb{1}(\cdot)$ denotes the indicator function, which equals 1 if the condition inside the parentheses holds true, and 0 otherwise.

- **Precision@k.** This setting is merely for purposes such as manually validating a fixed number of anomalies, where $Precision@k$ measures the relevance among our specified k samples. The mathematical expression is:

$$precision@k = \frac{\sum_{i \in top_k_indices} labels[i]}{k}, \quad (21)$$

where $top_k_indices$ is a collection of indices corresponding to the top k highest predicted values, identified using a fast partitioning algorithm. $labels$ refers to the array of true labels, and k denotes the number of top predicted values that are of interest to us.

- **Average precision**, which averages the precision-recall data into a single value. First, we need to compute the Precision-Recall (PR) curve. The definition of $precision@k$ is shown in Eq. (21). The definition of $recall@k$ can then be expressed as:

$$recall@k = \frac{\sum_{i \in top_k_indices} labels[i]}{\text{Total positive examples}}. \quad (22)$$

Next, we compute the Average Precision (AP). Typically, the area under the PR curve is approximated by discretizing the PR curve, which sums instead of integrating to approximate the area under the PR curve.

$$AP \approx \sum_k (recall@k - recall@(k-1)) \cdot precision@k, \quad (23)$$

where k iterates over all non-decreasing points on the PR curve (points where recall does not decrease). Therefore, the AP function computes the Average Precision using the aforementioned process, which evaluates the balance between precision and recall when predicting positives.

C. Implementation Details

The hyperparameter settings for validating our VSD-AD model on simulation data are presented in Table IV. The parameters are divided into three parts according to the process: graph convolution, model, and training parameters. Notably, the parameter δ_x representing the distance threshold between neighbors differs for detecting anomalous vehicles and anomalous scenarios, being 0.1 miles and 0.15 miles, respectively. Based on $\mathcal{W}$ and $sample_time$, the graph depth is determined to be 15. In the model parameter settings, the recommended calculation method for NFQ embedding space is $K = 2^{d_h}$, and we adopt a slightly larger vector space to better aid quantification. The $aggre_list$ indicates a combination of masked min and mean aggregators for the MMA aggregator, the rationale for which is detailed in Section III-C. It's worth mentioning that, in the Training section, lr represents the initial learning rate, which is halved every 50 epochs. Gradient clipping is used to prevent gradient explosion, with a maximum norm of 1.

For the HighD dataset, the window size $\mathcal{W}$ is set to 10s, leading to a graph depth of 10. Apart from this, we also adjust

TABLE IV
HYPERPARAMETER SETTINGS ON THE VSD-AD
MODEL USING SIMULATION DATASET

process	hyper parameter	description	value
Graph construction	δ_x	distance threshold	0.1/0.15miles
	δ_l	lane threshold	1lane
	$\mathcal{W}$	window size	15s
	$sample_time$	Sample interval	1s
Model	d_h	Hidden size	5
	K	embedding space	128
	$aggre_list$	aggregators	min, mean
	n_{hidden}	* $\mathcal{M}$ hidden size	9
Training	$dropout$	* $\mathcal{M}$ dropout	0.5
	$epochs$	Training epochs	300
	$batch_size$	Batch size	32
	lr	learning rate	0.05

the neighbor distance threshold δ_x to 0.2, and set the batch size to 128, with other parameters configured the same as in the simulation dataset.

Given that our proposed VSD-AD model requires a constant number of vehicles in each time window, many vehicle trajectories are not complete, such as those entering or leaving the observed road segment. Our approach is as follows: For the TransModeler Simulation dataset, owing to its large volume and controllable data source, we directly discard all incomplete vehicle trajectories. For the real-world HighD dataset, due to its uncontrollable data source, containing numerous incomplete segments and shorter sampling intervals, we complete the incomplete vehicle trajectories through linear extrapolation, assuming a constant vehicle speed for the extrapolated parts. These are masked during loss computation to prevent them from influencing the experimental results.

D. Results and Analysis

In this section, we conduct a quantitative analysis and compare the performance of our model on TransModeler Simulation data with baselines across different scenarios. We also perform ablation studies and hyperparameter analyses to understand the impact of each model component and the variations in hyperparameters on performance.

1) **Model Comparison:** In this section, we not only compare different anomaly vehicle detection methods in comprehensive scenarios but also evaluate the performance of anomaly scene detection against baselines. Subsequently, we compare the performance of anomaly vehicle detection in other scenarios. All baseline methods utilize the formula 18 and 19 to compute anomaly scores.

Initially, we conduct benchmark tests on the validation set, where all anomalous social dynamics are manifested on highways. The comparative experimental results for anomaly vehicle detection can be seen in Table V. Apart from DSAB, which captures inter-node correlations through graph attention mechanisms, our method is the only one capable of identifying specific behaviors, such as anomalous vehicles that deviate from typical social dynamics. Concerning precision across different k values, most methods do not exceed 0.6 and generally fall within the range of 0.1-0.3. This is because the social relationships between vehicles cannot be identified through simple mathematical methods or neural networks

TABLE V
ABNORMAL VEHICLE DETECTION PERFORMANCE ON THE
TEST SET OF COMPREHENSIVE SCENARIO

	Pre@100	Pre@200	Pre@500	Avg Pre	AUC
LTI	0.21	0.205	0.21	0.093	0.623
CVM	0.15	0.14	0.142	0.089	0.617
RTR	0.16	0.19	0.228	0.109	0.598
Seq2Seq	0.56	0.435	0.31	0.132	0.77
STGAE-biv	0.18	0.18	0.178	0.093	0.641
DSAB	0.82	0.775	0.726	0.381	0.9
VSD-AD(ours)	0.86	0.786	0.714	0.387	0.918

TABLE VI
ABNORMAL SCENE DETECTION PERFORMANCE ON THE
TEST SET OF COMPREHENSIVE SCENARIO

	Pre@100	Pre@200	Pre@500	Avg Pre	AUC
LTI	0.89	0.86	0.834	0.69	0.726
CVM	0.67	0.625	0.64	0.662	0.714
RTR	0.89	0.895	0.864	0.769	0.744
Seq2Seq	0.87	0.87	0.86	0.742	0.745
STGAE-biv	0.82	0.74	0.776	0.666	0.678
DSAB	0.93	0.95	0.912	0.859	0.841
VSD-AD(ours)	1	0.987	0.943	0.871	0.89

modeling individual vehicles. This microscopic approach to anomaly vehicle detection highlights this fact. In contrast, our method achieves a precision of over 0.7, even approaching 0.9 at its peak, where DSAB and our method are roughly comparable. Moreover, our average precision surpasses that of most methods by approximately 0.6%, and similarly, our ROC-AUC score is approximately 0.18% higher. These experimental results demonstrate that our VSD-AD method can delve deeper into the neighbor relationships between vehicles. While STGAE-biv and DSAB can also identify social relationships between vehicles, the former fails to capture lane-changing behaviors well due to its use of bivariate Gaussian trajectory loss, and the latter does not fully explore neighbor relationships. Not to mention that baselines like Seq2Seq, RTR, CVM, and LTI are unable to identify dynamic social relationships at all.

Furthermore, we conducted comparative experiments on the performance of anomaly scene detection among all methods, as shown in Table VI. Since anomaly scene detection involves macroscopically identifying abnormal patterns in entire scene segments, most methods perform well in this aspect. For instance, sudden deceleration, braking, or unreasonable acceleration of anomalous vehicles in specific scenes can be easily identified by simple non-parametric methods like LTI and CVM. Moreover, when anomalous vehicles are present in a scene, the scene's anomaly score is significantly higher compared to normal patterns.

Nevertheless, our method still maintains a leading position among all methods, with approximately a 2% improvement in precision and ROC-AUC indicators. This is noteworthy, considering that anomaly scene detection is generally well-addressed by existing methods due to its reliance on macro-level recognition of abnormal patterns in entire scene segments, unlike the more nuanced anomaly vehicle detection.

Next, we compared the performance between all baselines and VSD-AD in other distinct scenarios, including slow, speeding, tailgating, and stalled. The results can be referenced

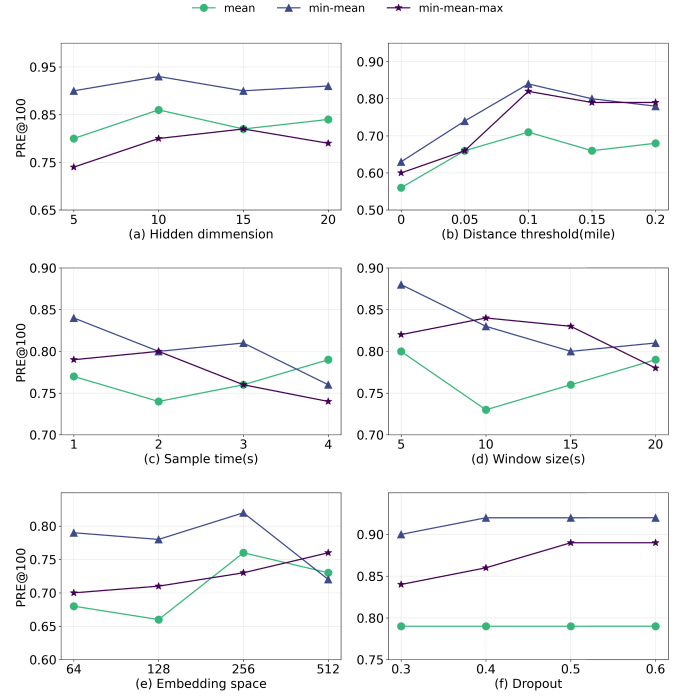


Fig. 4. Influence of different parameters. The model tends to favor the min-mean aggregator for fitting the data. Regarding the size of the code book embedding space, larger spaces enhance model accuracy but must be carefully balanced to avoid weakening quantization capabilities. The model exhibits insensitivity towards dropout and sampling time, affording us greater freedom in parameter selection.

in Table VII. From the metrics of pre@100 and Ave Pre, it can be observed that for slow and stalled vehicles, most methods fail to recognize their patterns. Only by deeply modeling the social relationships between vehicles can their anomalies be effectively detected. In these two scenarios, our method achieves optimal performance, with approximately a 1% improvement in each metric compared to the second-highest method.

It is evident that almost all methods, including the non-parametric approach CVM specifically designed for speeding and tailgating scenarios, perform well in the speeding scenario. Although DSAB stands out as the most outstanding, almost unbeatable, reaching an impressive 0.995 at ROC-AUC, the performance gap of our model remains within an acceptable range. Given that CVM follows different car-following dynamics in its initial setup, its performance surpassing all methods in the tailgating scenario is not surprising. However, it is worth noting that no method outperforms ours on every metric. we consistently outperform others in most scenarios.

2) *Hyper Parameter Analysis*: In this section, we explore the impact of different hyperparameter configurations on the model, considering the customization capability of our proposed model. We employ different GRU-built graph convolution modules to validate the model's sensitivity to them. The RMMA (Recurrent Multi-head Message Aggregator) built-in aggregator combinations used are min, min-mean, and min-mean-max.

Firstly, we investigate the sensitivity of the model to the hidden dimension during the model and social dynamics relationship graph construction process. From Figure 4a, it can

TABLE VII
PERFORMANCE OF ABNORMAL VEHICLE DETECTION IN DIFFERENT INDEPENDENT SCENARIOS

	Slow			Speeding			Tailgating			Stalled		
	Pre@100	Ave Pre	AUC	Pre@100	Ave Pre	AUC	Pre@100	Ave Pre	AUC	Pre@100	Ave Pre	AUC
LTI	0	0.023	0.443	0.82	0.512	0.877	0.02	0.161	0.835	0	0.03	0.01
CVM	0.01	0.023	0.432	0.67	0.472	0.871	0.06	0.177	0.84	0	0.03	0.01
RTR	0	0.03	0.466	0.61	0.365	0.655	0.08	0.102	0.4	0	0.046	0.41
Seq2Seq	0.02	0.046	0.668	0.85	0.549	0.923	0.18	0.091	0.709	0.03	0.056	0.676
STGAE-biv	0.01	0.048	0.573	0.83	0.335	0.712	0.01	0.055	0.491	0	0.025	0.454
DSAB	0.97	0.246	0.841	1	0.907	0.995	0.33	0.148	0.747	0.98	0.148	0.762
VSD-AD(ours)	0.98	0.261	0.89	1	0.894	0.976	0.39	0.162	0.767	0.98	0.15	0.769

be observed that the model is not sensitive to the hidden dimension, and there is no discernible pattern. To ensure fewer model parameters, we choose 5 as the hidden dimension. Next, by observing Figures 4b, 4c, and 4d, the distance threshold signifies that vehicles within this threshold are defined as neighbor nodes during the construction of the spatiotemporal relationship graph. When the value is 0, the model focuses only on a single vehicle without considering social dynamics. From Figure 4b, it is evident that the model's performance is optimal when the distance is 0.1.

Considering the time dimension, we initially set the sampling interval to 1 second and change the window size from 5 seconds to 20 seconds. Then, we fix the window size at 15 seconds and change the sampling interval from 1 second to 4 seconds. Figures 4c and 4d show that different window sizes and sampling frequencies have minimal impact on the overall performance of the model. The model achieves optimal performance when the embedding space for RMMA is set to 256, and it is sensitive to this parameter according to 4e.

Finally, considering the introduced NFQ module, Figure 4f compares the impact of its parameter settings. It can be observed that after reaching an optimal solution, further growth in the dropout does not contribute to improving the model.

Each line plot in Figure 4 illustrates the impact of different RMMA built-in aggregator combinations under various parameters. In summary, the min-mean aggregator combination is most suitable for fitting the model to the simulated dataset. It is essential to note that aggregators like min, mean, and max do not indicate the nearest, average, and farthest distances in the spatiotemporal vehicular social dynamics graph, but rather their impact on ego-vehicles as indicated by the masked representations learned by the MLP formula.

3) *Ablation Study*: In this study, we aim to evaluate the impact of specific modules within the proposed VSD-AD model on the overall model performance by selectively removing them. We objectively assess the influence of the MMA (Multi-Mask Aggregator) module and the NFQ (Neighbor Fusion Queue) module on the model using the AUC (Area Under the Curve) metric on a simulated dataset across four scenarios (Slow, Speeding, Tailgating, Stalled). Notably, the MMA module utilizes a min-mean aggregator combination.

As depicted in Figure 5, the removal of the MMA module, replaced with GCN (Graph Convolutional Network) for graph convolution operations to process neighboring vehicle information, results in a significant performance decrease.

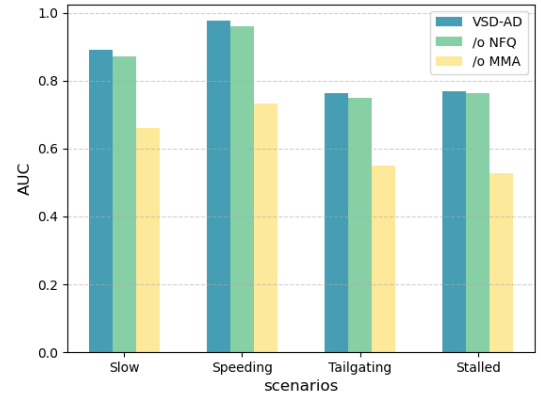


Fig. 5. The ablation study reveals that the MMA module significantly enhances the overall performance of the model, whereas the NFQ module, while contributing to improvements, functions more akin to fine-tuning the model.

Across each scenario, there is an approximate 0.26 drop in accuracy. Conversely, solely eliminating the NFQ module, typically resulting in a 1%-2% difference in accuracy.

E. Qualitative Results

In this section, we qualitatively studied the performance of our proposed VSD-AD model on high-definition data captured by drones in the real world. Unlike simulated data, the HighD dataset has relatively fewer trajectories and different road condition patterns. However, our model still manages to capture these patterns and detect anomalies.

The data in HighD primarily originates from German highways, where speed limit policies are relatively lenient, with some areas even having unrestricted speeds. Figure 6a illustrates the road conditions during normal driving. Calculated from the training set, the average speeds from right to left lanes are 56, 69, and 76 mph, respectively. Similar to driving rules in China, faster vehicles tend to be closer to the left, while slower ones are inclined to be on the right. Therefore, judging a vehicle's abnormality solely based on whether its speed exceeds the limit is insufficient. It's also necessary to consider if its speed aligns with other vehicles in the same lane. Otherwise, it should be considered abnormal. In this section, we manually inspected the segments where the vehicles with the highest anomaly scores were found. We discovered that the abnormal vehicles fell into the two categories mentioned earlier: speeding/slowing down, indicating aggressive driving with excessive/insufficient acceleration and speeds not consistent with the normal range of the lane.

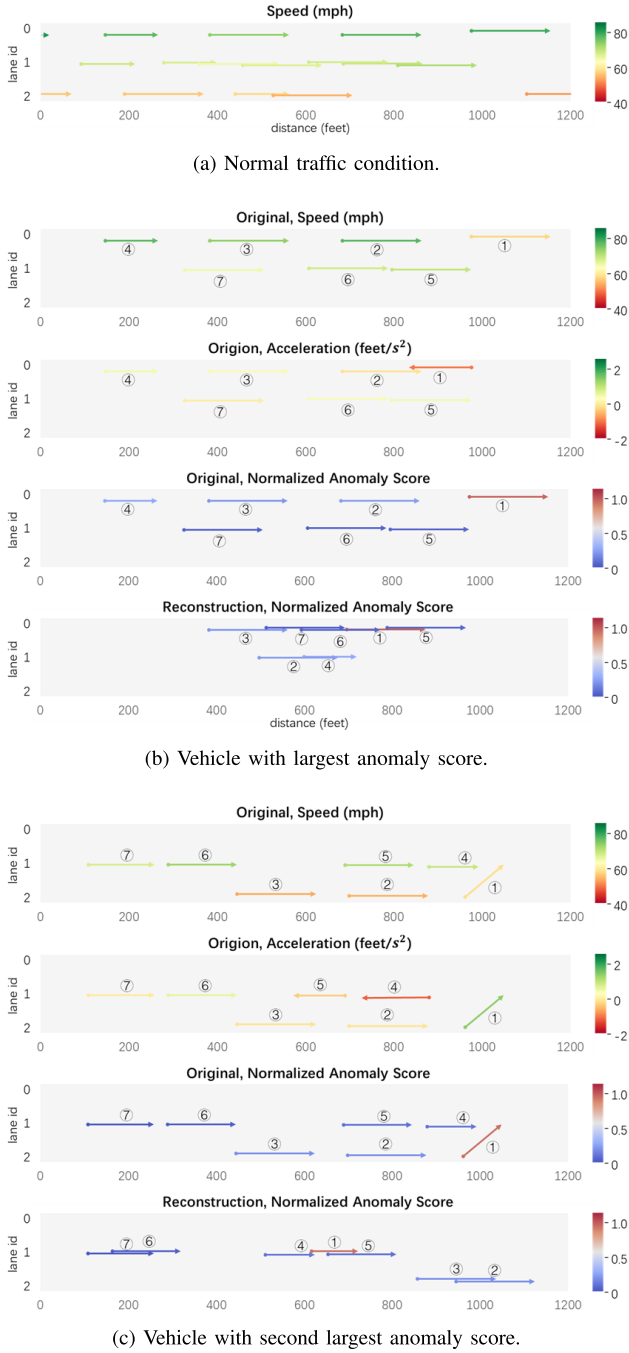


Fig. 6. Qualitative results. The visualization demonstrates the detection of the top two instances with the highest anomaly scores by the model. The reconstructed scenes appear significantly different from the original scenes, yet the model successfully discerns abnormal and suspicious vehicles within them.

We visualize our experimental results in Figure 6, showcasing the top two scenes with the highest anomaly scores from four perspectives: speed, acceleration, normalized anomaly scores, and normalized anomaly scores reconstructed by the model. Each arrowhead dot represents the starting point, while the triangle represents the end point. In the acceleration view that is shown in Figure 6b, arrows pointing in the opposite direction indicate vehicles are decelerating. In Figure 6b, vehicle number 1 suddenly decelerates without any interference from vehicles ahead. In Figure 6c, vehicle number 1 attempts to accelerate from lane 2 to lane 1, which requires a higher

speed, causing vehicle number 4 to brake urgently. Our model accurately labels vehicle number 4 as normal in this scenario.

V. CONCLUSION

In this study, we address the critical task of detecting anomalous social dynamics using trajectories obtained from a high-speed highway video surveillance system supported by the Internet of Things. Our approach focuses on employing VSD-AD, an innovative autoencoder supported by recurrent multi-mask aggregators, to precisely identify anomalous vehicles. By leveraging this method, we successfully capture the complex spatiotemporal dynamics of trajectory patterns, taking into account the interactions between neighboring vehicles and the inherent randomness in driving behavior. Through comprehensive experiments on both simulated and real-world datasets, our model demonstrates its scalability, making it suitable for large-scale highway monitoring systems with thousands of vehicles. Furthermore, our user-customizable approach exhibits significant efficacy in detecting various anomalous scenarios. Particularly noteworthy is its outstanding performance in identifying anomalies in individual vehicles, placing it at the forefront of anomaly detection methods. This implies its potential to accurately pinpoint specific problematic vehicles within massive traffic flows, thereby promoting enhanced traffic management and safety protocols.

Regarding the shortcomings and perspectives of this study, it is notable that the MMA modules incur significant computational costs, requiring several hours for training in our experimental setup. Despite achieving commendable results in both simulation and real-world testing, there remains a pressing need to enhance the algorithm's time complexity in the immediate future. Moreover, the VSD-AD model operates on a data-driven basis, necessitating extensive data acquisition and preprocessing efforts, particularly challenging in obtaining highway data, which demands substantial costs, time, and manpower. To address this challenge, we anticipate that advancements can swiftly be made through temporal data reconstruction techniques or the generation of highway imagery via visual direction neural networks and data augmentation solutions [56], thereby augmenting the quantity of authentic data and reducing deployment costs for the model.

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